



Migrating to Proxmox with Shift Toolkit

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1 Introduction

In this lab, you will explore the Migration feature of the NetApp Shift Toolkit. You will perform a Migration, which moves your VM from VMware ESXi to Proxmox by converting VM files to the appropriate format and transferring them to the destination hypervisor.

Deciding on the best course of action involves weighing technical and commercial risks, such as migrating workload VMs to alternate hypervisors, focusing on business-defined objectives, and controlling vendor lock-in. This approach helps optimize licensing costs and allows IT budgets to be allocated to the right areas, rather than spending on unused cores for a specific hypervisor. However, migration time, associated downtime, and the availability of enterprise support for alternative platforms like Proxmox can be limiting factors for many teams.

The NetApp Shift toolkit makes migrating virtual machines (VMs) fast and efficient, eliminating many of these concerns. This standalone product enables seamless migration of VMs from VMware ESXi to Proxmox, and also supports disk-level conversions between various virtual disk formats. With the built-in capabilities provided by ONTAP, migrations can be completed quickly with minimal downtime.

One advantage of in-place conversion is that no data is moved through a third-party service—the change happens directly on the disk, on a clone of the original VM. A space-efficient clone preserves the integrity of the source while enabling the metadata updates needed to convert the virtual disk format. Hypervisor CPU and network performance are unaffected by this migration. This results in less downtime, especially when coordinating the migration of multiple VMs from VMware to Proxmox.

To further streamline operations between Proxmox Virtual Environment (PVE) and ONTAP storage, NetApp offers Storage Manager for Proxmox. This tool simplifies and enhances integration, making it easier to manage storage resources efficiently.

2 Lab Activities

The following lab activities demonstrate how to set up the NetApp Shift Toolkit, to Migrate, and to Convert VMDK to QCOW2 or RAW for use in a Proxmox VE environment.:

- [Shift Toolkit Overview](#)
- [Migrate Virtual Machines](#)
- [Convert Virtual Machines](#)

2.1 Shift Toolkit Overview

The NetApp Shift Toolkit is a powerful, easy-to-use solution designed to facilitate seamless migration of virtual machines (VMs) between different hypervisors. For organizations transitioning to or operating in a Proxmox Virtual Environment, Shift Toolkit provides an intuitive graphical user interface (GUI) that streamlines the process of moving VMs from platforms like VMware ESXi to Proxmox.

By leveraging NetApp’s advanced FlexClone technology, the Shift Toolkit rapidly converts virtual disk formats—such as VMDK to QCOW2—to ensure compatibility with Proxmox’s KVM-based infrastructure. In addition to disk conversion, the toolkit automates the creation and configuration of destination VMs, reducing manual intervention and minimizing downtime. It also tests the completed migration and reports the results back to the UI, providing clear confirmation of a successful migration.

With support for multi-hypervisor environments and bidirectional migration capabilities, NetApp Shift Toolkit empowers IT teams to efficiently manage and migrate workloads, making it an ideal solution for organizations seeking flexibility and simplicity in their virtualization strategy.

Shift toolkit provides flexibility in a multi-hypervisor environment by supporting bi-directional conversion between the following hypervisors:

- VMware ESXi to Proxmox

Shift toolkit supports disk-level conversions of virtual disks between hypervisors for the following disk format:

- VMDK # VHDX
- VHDX # VMDK
- VMDK # QCOW2
- VMDK # RAW

This section details the steps to create a source and destination target using the NetApp Shift Toolkit through the user interface.

Activities
Add Sites
Define a Resource Group
Develop a Blueprint

2.1.1 Add Sites

The first step involves discovering and adding the details of the source vCenter and the target Proxmox to the Shift Toolkit.

1. Click the **Shift Toolkit** icon.

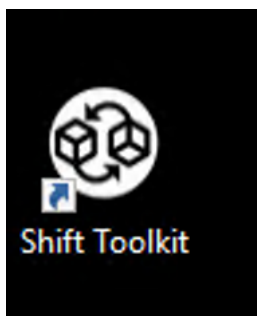


Figure 2-1:

2. Click **Advanced**.

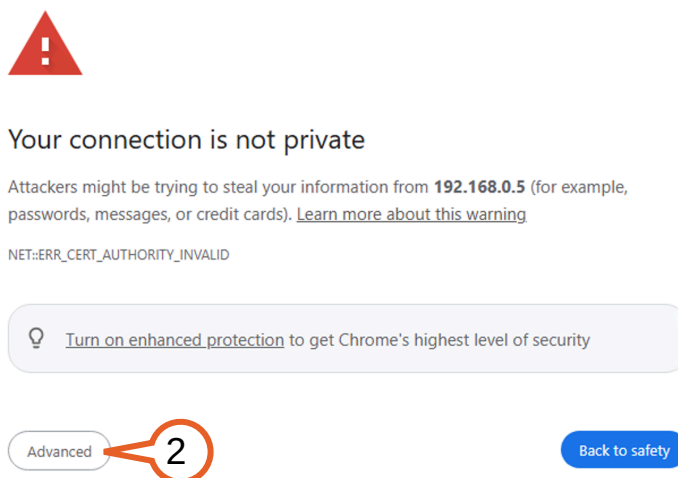


Figure 2-2:

3. Click **Proceed to 192.168.0.5 (unsafe)**.

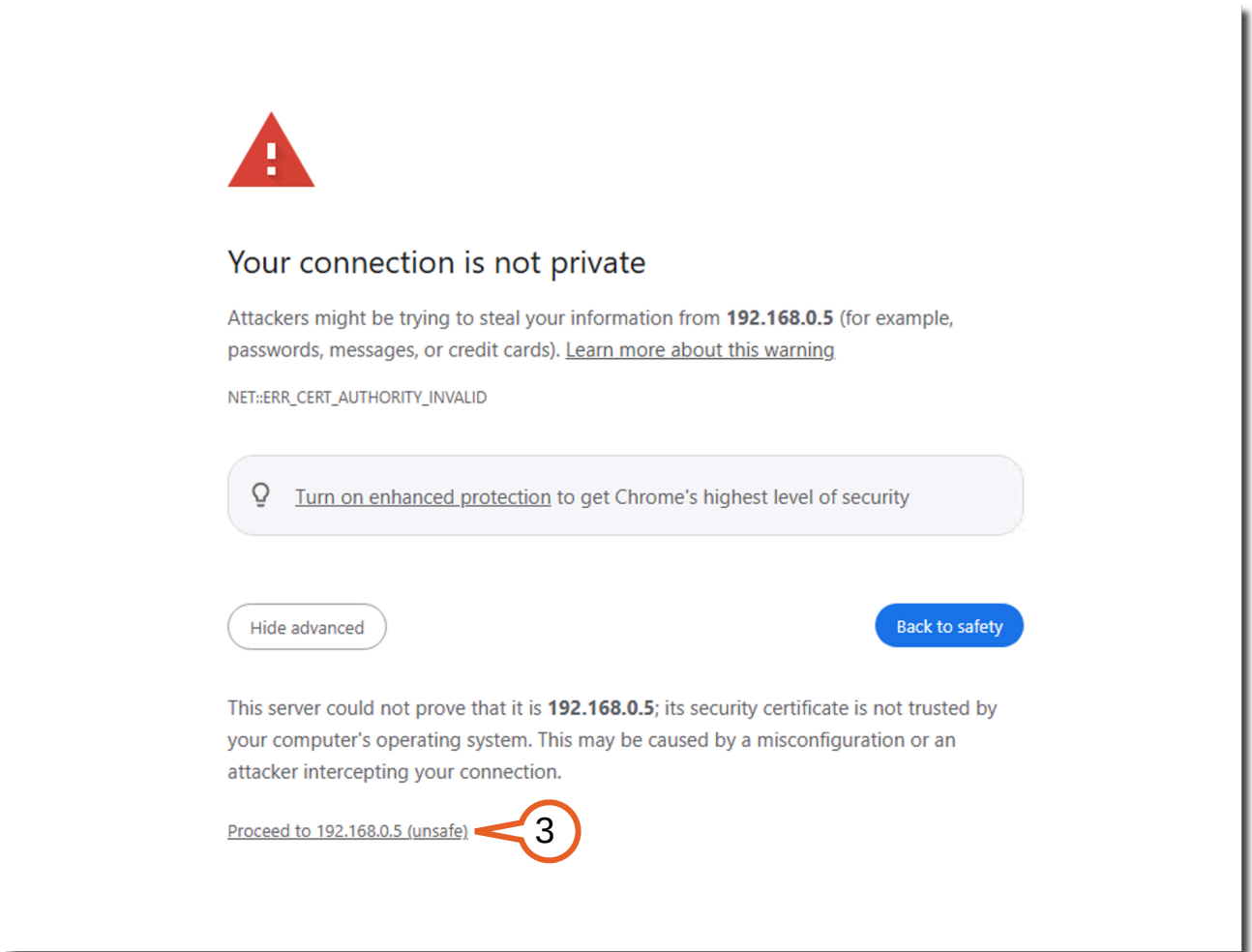
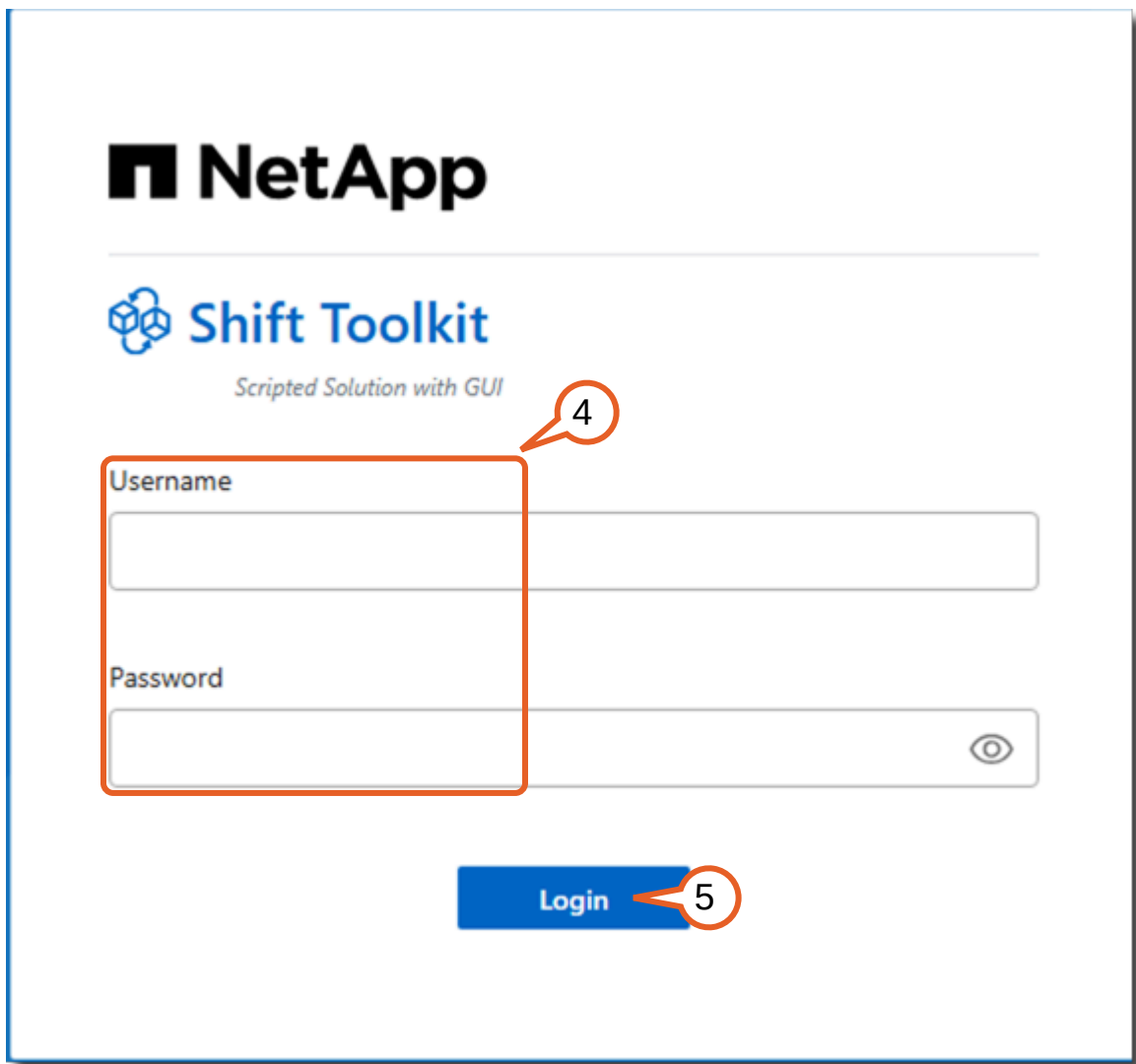


Figure 2-3:

4. Enter the following credentials:
 - Username: **admin**
 - Password: **Netapp1!**

5. Click **Login**.



The image shows the NetApp Shift Toolkit login interface. At the top is the NetApp logo, followed by the Shift Toolkit logo and the tagline "Scripted Solution with GUI". Below this are two input fields: "Username" and "Password". The "Username" field is highlighted with a red rectangle and a callout bubble containing the number "4". The "Password" field has a toggle icon on the right. Below the input fields is a blue "Login" button, which is highlighted with a red callout bubble containing the number "5".

Figure 2-4:

6. Click **Add Site**.

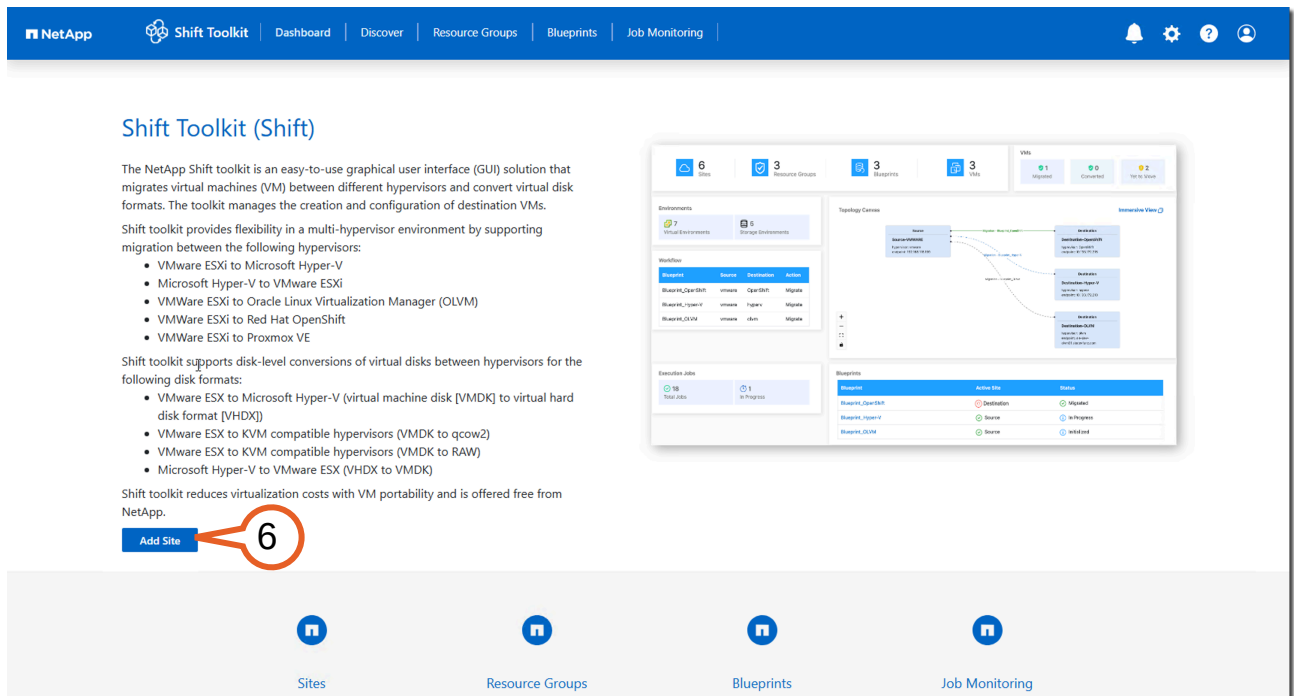


Figure 2-5:

7. Select **Source** for the new site you will create.

8. Click **Continue**.

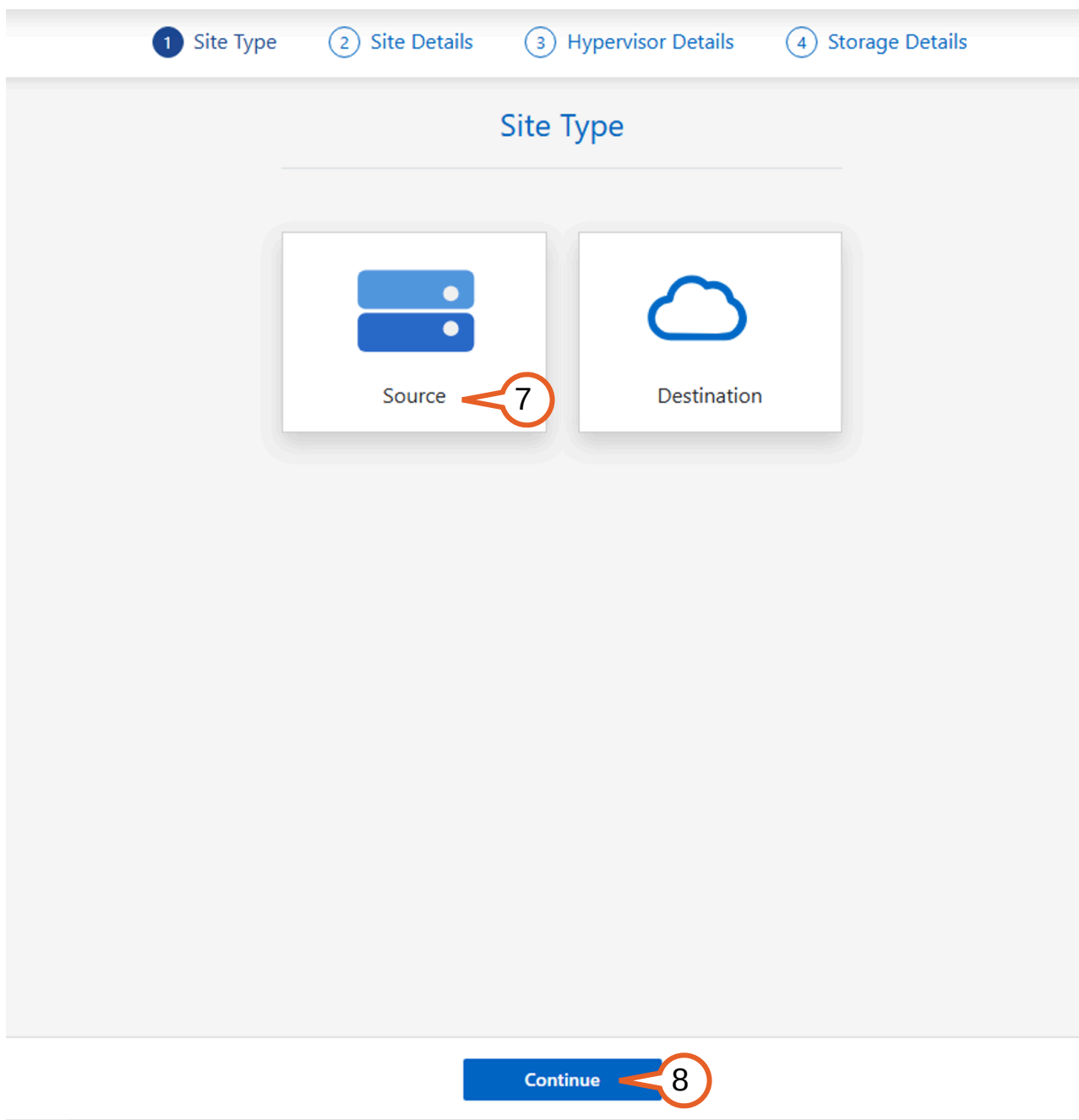


Figure 2-6:

9. Under Source Site Details page Site Name, enter **vc1**.

10. Select the following:

- Hypervisor: **VMware**
- Site Location: **On Prem**
- Connector: **default-connector**

11. Click **Continue**.

Source Site Details

9 Site Name
vc1

10 Hypervisor
VMware

Site Location
On Prem

Connector
default-connector

Previous Continue 11

Figure 2-7:

12. For Source vCenter Details, enter the following:
- vCenter Endpoint: **vc1.demo.netapp.com**
 - vCenter Username: **administrator@demo.local**
 - vCenter Password: **Netapp1!**
13. Select the check box for **Accept self-signed certificates**.

14. Click **Continue**.

The screenshot shows a web form titled "Source vCenter Details". It contains three input fields: "vCenter Endpoint" with the value "vc1.demo.netapp.com", "Enter vCenter username" with the value "administrator@demo.local", and "vCenter Password" with masked characters ".....". Each field has an information icon (i) to its right. A red box labeled "12" encloses these three fields. Below the fields is a checkbox labeled "Accept self-signed certificates" which is checked; a red box labeled "13" points to this checkbox. At the bottom of the form are two buttons: "Previous" and "Continue". A red box labeled "14" points to the "Continue" button.

Figure 2-8:

15. On Source Storage Details page enter the following:
- Storage Endpoint: **cluster1.demo.netapp.com**
 - Storage Username: **admin**
 - Storage Password: **Netapp1!**
16. Select the check box for **Accept self-signed certificate**.

17. Click **Create Site**.

✓ Site Type ✓ Site Details ✓ Hypervisor Details **4 Storage Details**

Source Storage Details

15

Storage Endpoint ⓘ
cluster1.demo.netapp.com

Storage Username ⓘ
admin

Storage Password ⓘ
..... ⓘ

16 ☒ Accept self-signed certificates

17

Previous **Create Site**

Figure 2-9:

18. Once created a notification will appear at the bottom of the screen, “Source Site is registered. Discovery Initiated!”.



Note: Once added, Shift toolkit will perform an automatic discovery and display the VMs along with the relevant metadata information. Shift toolkit will automatically detect the networks and port groups used by the VMs and will populate them

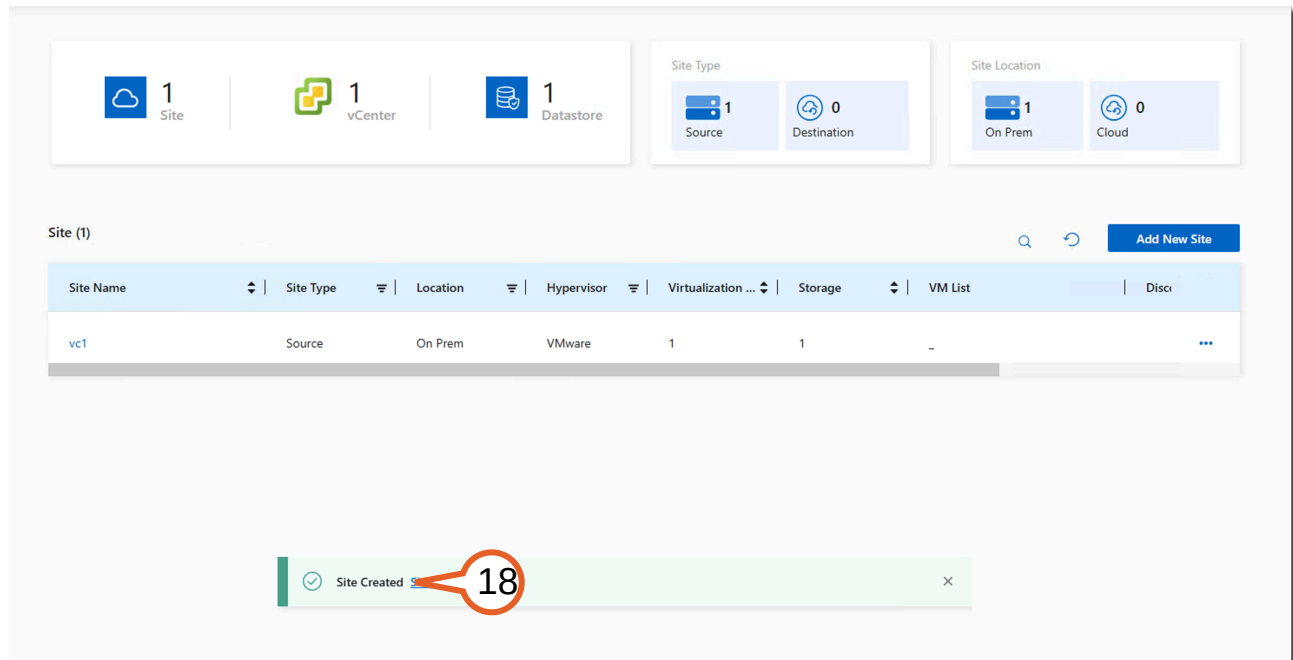


Figure 2-10:

19. Click **Add New Site**.

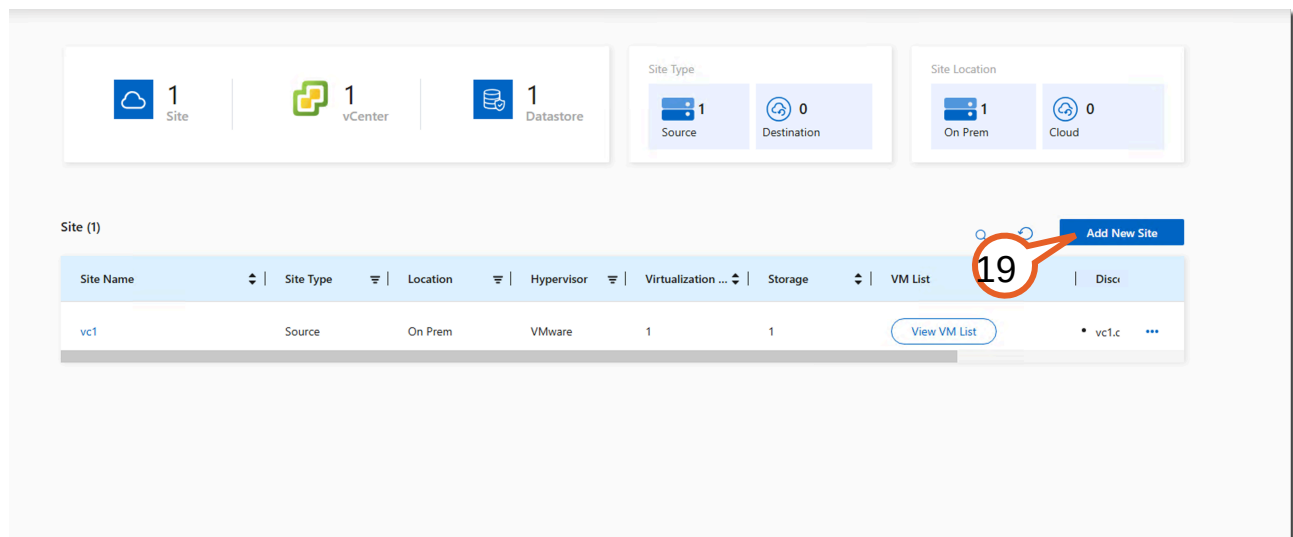


Figure 2-11:

20. Click **Destination**.

21. Click **Continue**.

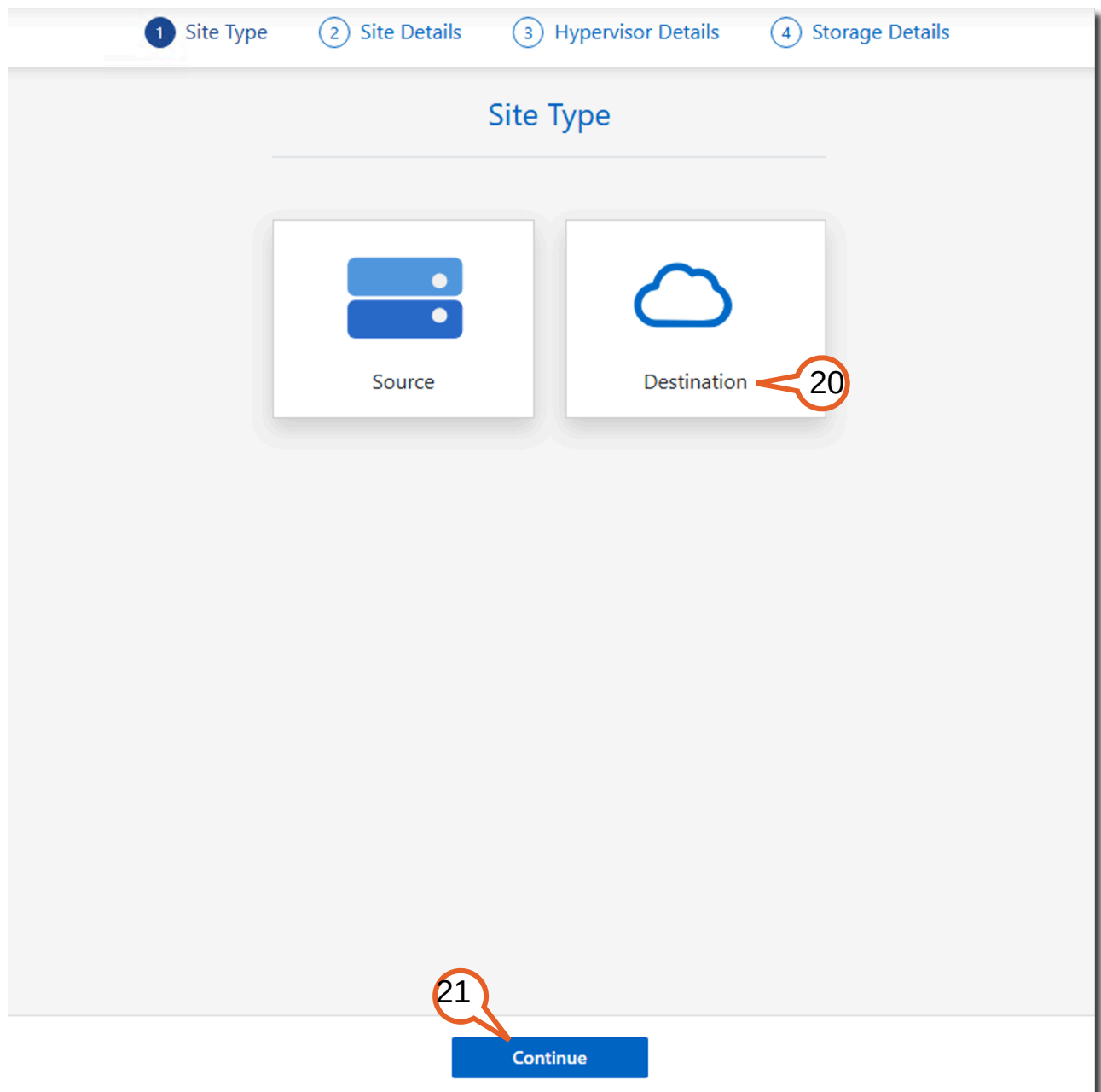


Figure 2-12:

22. On the Destination Site Detail page add the following:

- Site Name: **PVE**
- Hypervisor: **Proxmox VE**
- Site Location On Prem (default)
- Connector: **default-connector**

23. Click **Continue**.



Note: KVM is also listed as an option for a destination, but it is not covered in this lab at this time.

Site Type 2 Site Details 3 Hypervisor Details

Destination Site Details

22

Site Name *i*
PVE

Hypervisor *i*
Proxmox VE

Site Location *i*
On Prem

Connector *i*
default-connector

Previous Continue 23

Figure 2-13:

24. In Destination Proxmox Details, add the following details:

- Proxmox Endpoint: **192.168.0.91:8006**
- Username: **root**
- Password: **Netapp1!**

25. Click the checkbox for **Accept self-signed certificates**.

26. Click **Create Sites**.

The screenshot shows the 'Destination Proxmox Details' form in the 'Hypervisor Details' step. The form is titled 'Destination Proxmox Details' and has three tabs: 'Site Type', 'Site Details', and 'Hypervisor Details' (the active tab). The form contains the following fields:

- Proxmox Endpoint:** A text input field containing '192.168.0.91:8006'. This field is highlighted by callout box 24.
- Username:** A text input field containing 'root'.
- Password:** A password input field with masked characters '.....' and a toggle icon (an eye) to the right.
- Accept self-signed certificates:** A checkbox that is checked, with the text 'Accept self-signed certificates' to its right. This checkbox is highlighted by callout box 25.

At the bottom of the form, there are two buttons: 'Previous' and 'Create Site'. The 'Create Site' button is highlighted by callout box 26.

Figure 2-14:

27. Once created a notification will appear at the bottom of the screen that states “Site Created”.

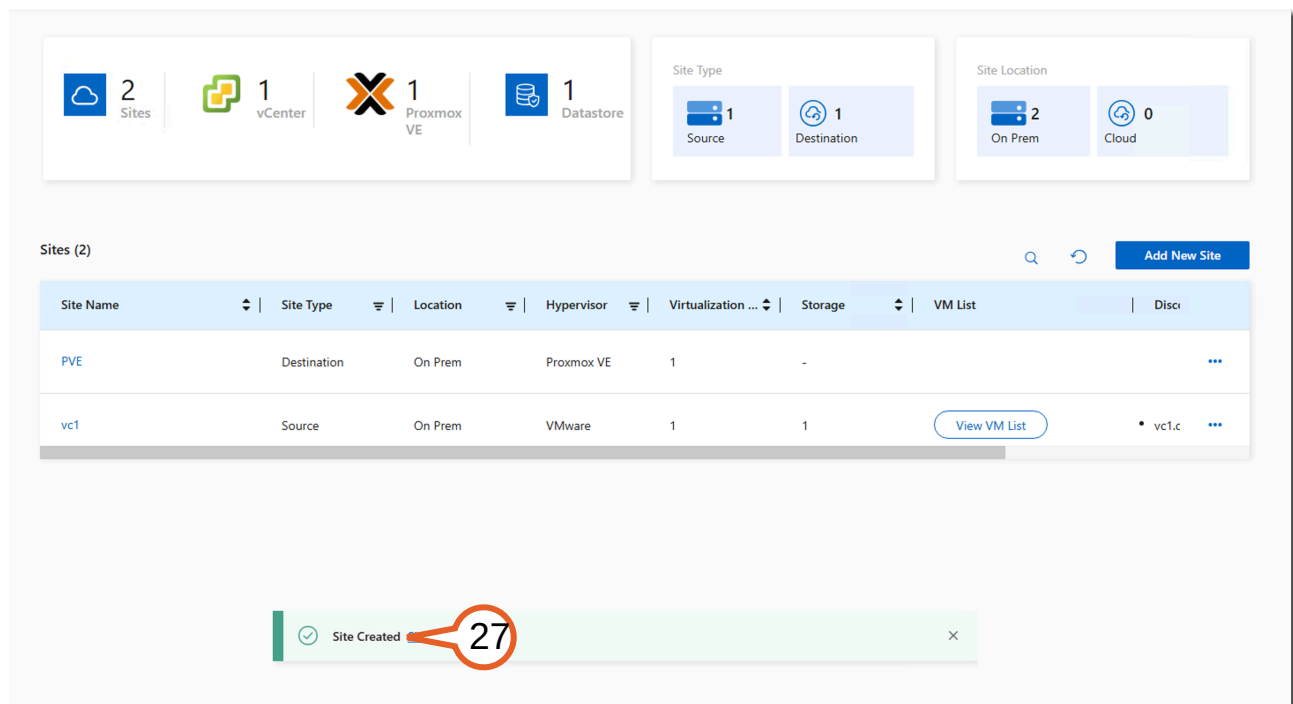


Figure 2-15:

You have now created a source and destination site to specify Proxmox as the destination hypervisor and VMware as your source. Next you will define resource groups.

2.1.2 Define a Resource Group

A resource group is a logical container for virtual machines (VMs) that you want to migrate or convert from one platform to another. Shift Toolkit resource groups allow you to organize sets of dependent VMs into logical groups, preserving their boot order and boot delay configurations.

In this activity, you create a resource group to contain three VMs supporting the "app1" application..

1. Click **Resources Groups**.

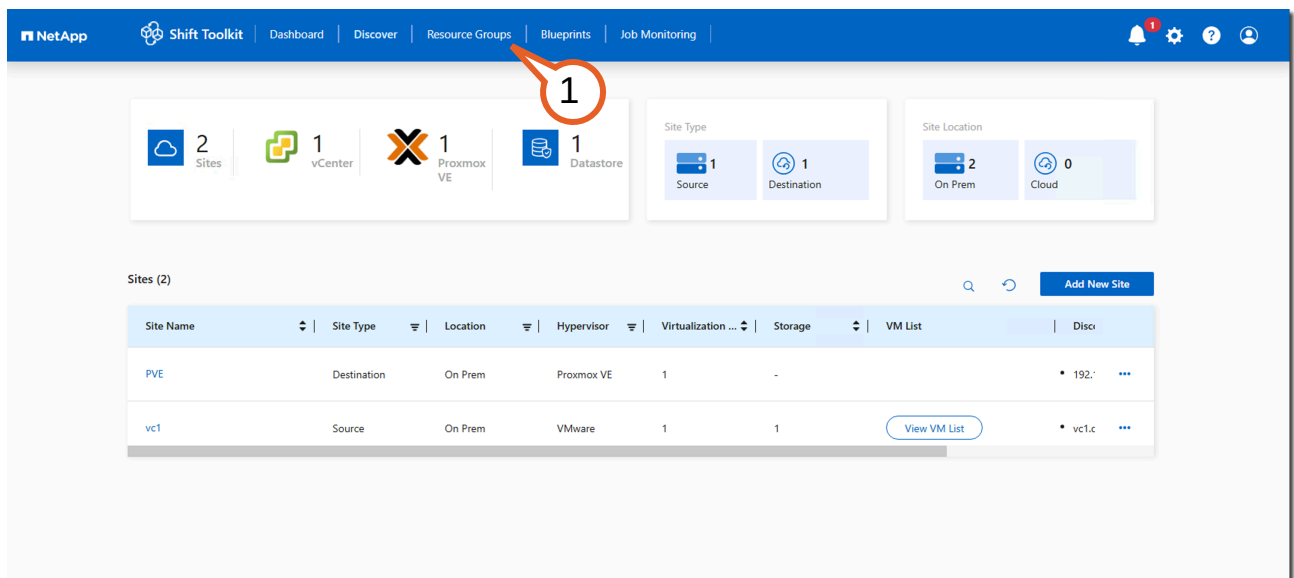


Figure 2-16:

2. Click **Create New Resource Group**.

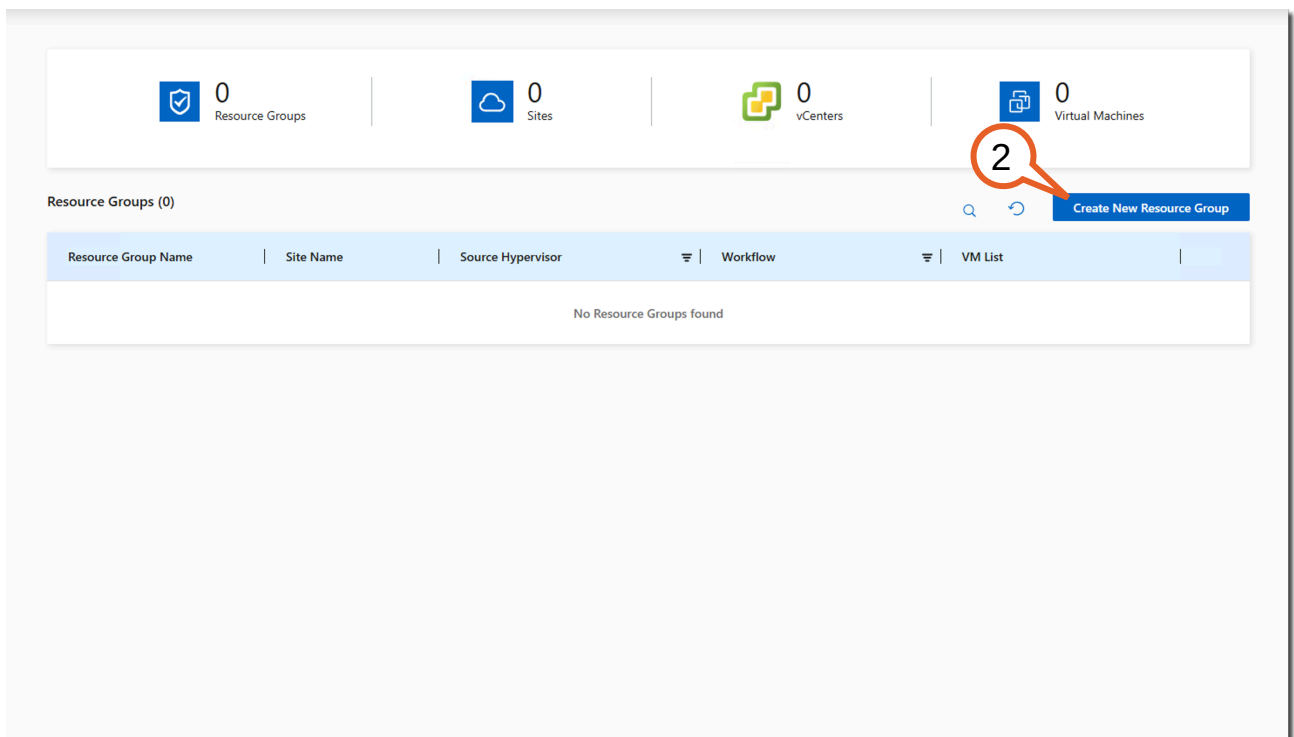


Figure 2-17:

3. Select **vc1**.

4. Click **Create**.

New Resource Group

Select Source Site

vc1

Create

Figure 2-18:

5. Add the following configuration Resource Group Details:
 - Resource Group Name: **Rg -vc1**
 - Associated Site: **vc1**(default)
 - Associated vCenter: **vc1.demo.netapp.com**(default)
 - Destination Site: **PVE**

6. Click **Next**.

< 1 Resource Group Details 2 Select Virtual Machines 3 Destination Details >

Resource Group Details

5

Resource Group Name

RG-vc1

Associated Site *i*

vc1

Associated Site vCenter *i*

vc1.demo.netapp.com

Destination Site *i*

PVE

Workflow *i*

Clone based Migration

Next 6

Figure 2-19:

7. On the Select Virtual Machine page, select datastore **nfs1**.

8. Select the following vms:

- **app1-db**
- **app1-api**
- **app1-web**

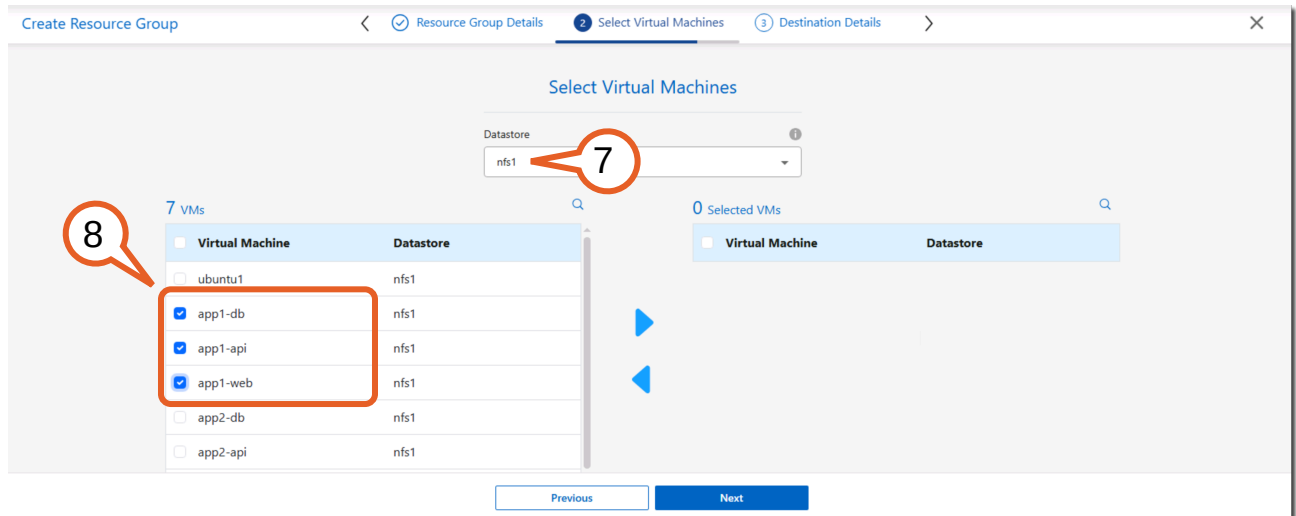


Figure 2-20:

9. Click the **arrow** to move the following vms to the Selected VMs column.

10. Click **Next**.

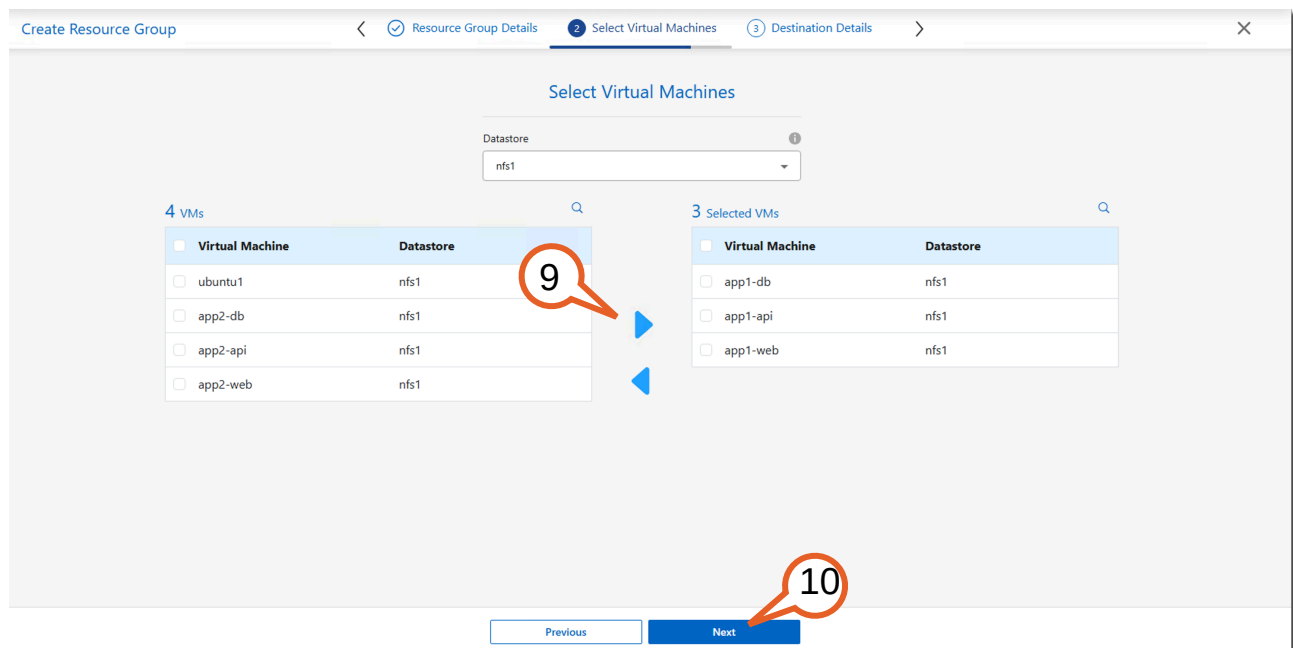


Figure 2-21:

11. For Migration Details select the following:

- Destination Site: **Proxmox**
- Destination Proxmox: **192.168.0.91:8006**
- ONTAP Volume nfs1 (default)

12. For Datastore Qtree Mapping select the following:

- Source: nfs1
- Destination Qtree: **PVE**



Note: More information about the qtree requirements are located here: <https://docs.netapp.com/us-en/netapp-solutions-virtualization/migration/shift-toolkit-install-prereqs.html>

13. Click **Next**.

Figure 2-22:

14. Click **Create**.

Create Resource Group

Boot order and Delay

3 VMs

VM Name	Boot Order	Boot Delay (secs)
app1-web	3	0
app1-api	3	0
app1-db	3	0

Previous Create

Figure 2-23:

15. Once created, a notification will appear at the bottom of the screen that states “Resource group registered”.

1 Resource Group | 1 Site | 1 vCenter | 3 Virtual Machines

Resource Group (1)

Resource Group Name	Site Name	Source Hypervisor	Workflow	VM List
RG-vc1	vc1	VCenter (vc1.demo.netapp.com)	Migration	View VM List

Resource Group created successfully

Figure 2-24:

Now that the VM is assigned to resource group, the next activity will show you how to develop a migration blueprint.

2.1.3 Develop a Blueprint

In this activity, you create a blueprint plan to prepare for a virtual machine migration.

1. Click **Blueprints**.

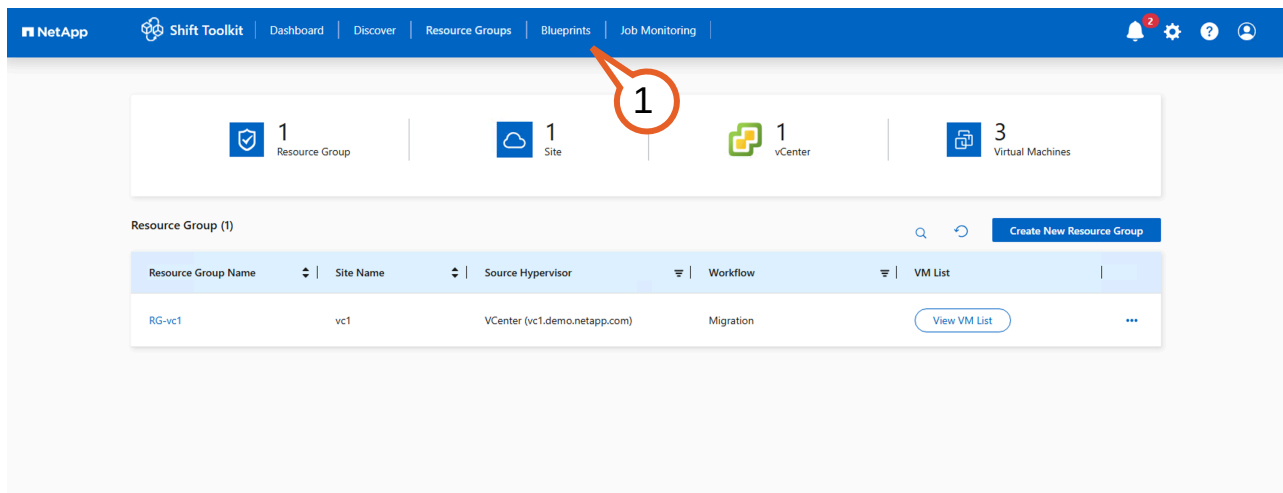


Figure 2-25:

2. Click the **Create New Blueprint** drop-down menu .
3. Select **Create New Blueprint**.

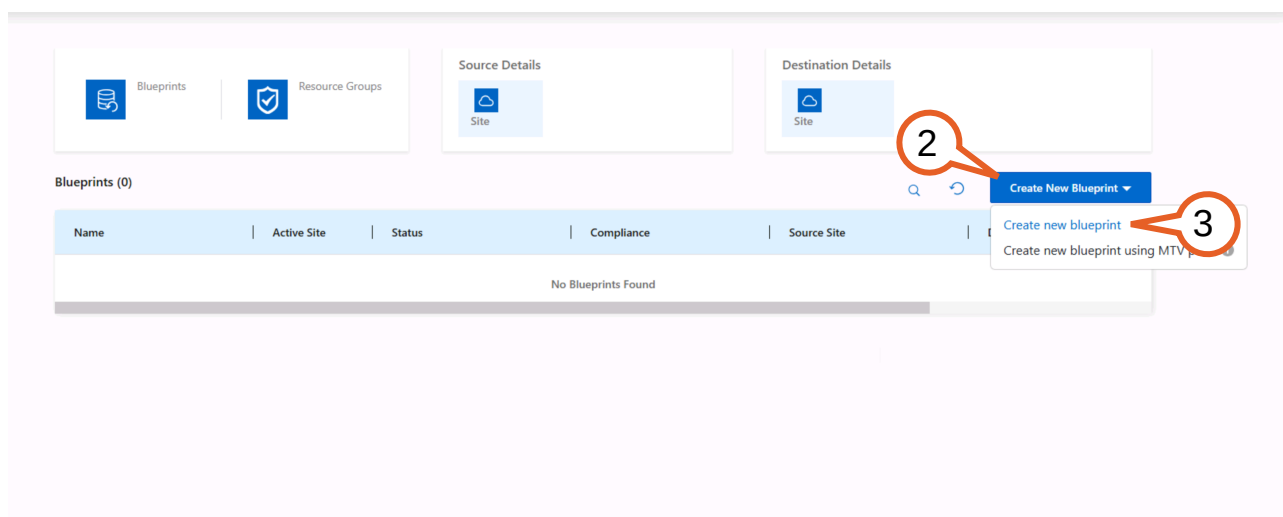


Figure 2-26:

4. Under Blueprint Details add the following:



Tip: When creating a blueprint in Shift Toolkit including multiple resource groups (RGs) within the same blueprint. This approach allows you to prepare and coordinate the migration of all related systems as a specific batch, ensuring dependencies and boot sequences are maintained across multiple VMs during the migration process. This helps streamline large-scale migrations and reduces the risk of missing critical relationships between systems.

- Blueprint Name: **Migrate-vc1**

5. For Resource Mapping, select the following:

- Source Site: **vc1**
- Destination Site: **Proxmox**

6. For Cluster and Host Mapping select the following:

- Source Site Resource: **vc1-cluster**
- Destination Site Resource: **PVE**

7. Click **Add**.

8. Click **Continue**.

The screenshot shows the 'Blueprint Details' configuration page with the following elements and callouts:

- 4**: Callout to the 'Blueprint Name' input field containing 'Migrate-vc1'.
- 5**: Callout to the 'Resource Mapping' section header.
- 6**: Callout to the 'Cluster and Host Mapping' section header.
- 7**: Callout to the 'Add' button in the 'Cluster and Host Mapping' section.
- 8**: Callout to the 'Continue' button at the bottom of the page.

The 'Resource Mapping' section contains the following fields:

Source Site	Destination Site
vc1	PVE

Source vCenter	Destination Proxmox VE
vc1.demo.netapp.com	192.168.0.91:8006

The 'Cluster and Host Mapping' section contains the following fields:

Source Site Resource	Destination Site Resource
Cluster1	pve-1

Below the mapping fields is a table with the following structure:

Source Resource	Destination Resource
No Mappings added!	

Figure 2-27:

9. Under Select Resource Group, double-click **RG-vc1** to the Selected Resource Group.

10. Click **Continue**.

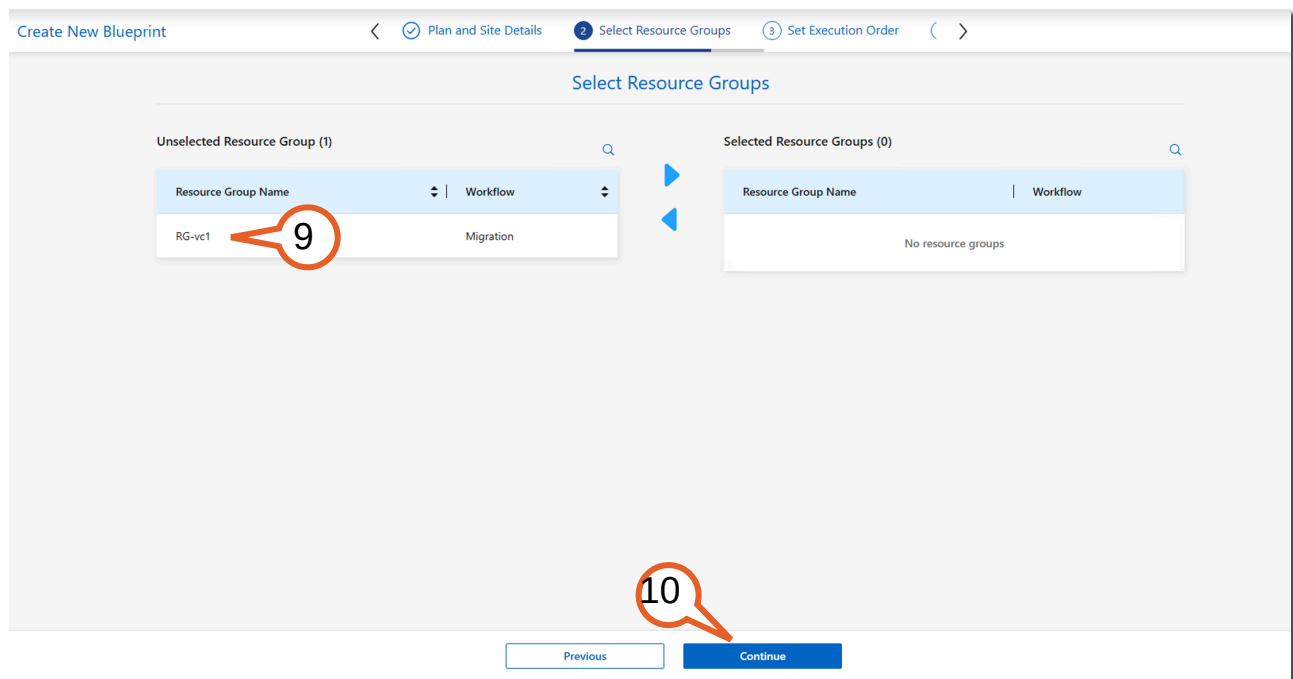


Figure 2-28:

11. Under Network Mapping, click **Add**.

12. Click **Continue**.

The screenshot shows the 'Migration Details' page in the NetApp Shift Toolkit. The navigation bar at the top indicates the current step is '3 Set Execution Order'. The 'Select Execution Order' section shows a table with 'RG-vc1' as the Resource Group Name and '3' as the Execution Order. Below this is the 'Network Mapping' section, which has two tabs: 'Target' and 'Test'. Under the 'Target' tab, there are two dropdown menus: 'Source Site Resource' (set to 'DPortGroup0') and 'Destination Site Resource' (set to 'isolated'). An 'Add' button is to the right of these dropdowns, circled with a red '11'. Below the dropdowns is a table with columns 'Source Resource' and 'Destination Resource', which is currently empty with the text 'No network mappings added!'. At the bottom of the page, there are 'Previous' and 'Continue' buttons. The 'Continue' button is circled with a red '12'.

Figure 2-29:

13. In the Service Account section, add the following credentials for the Linux machine:

- Linux Username: **root**
- Password **Netapp1!**

14. Click **Apply To All**.

15. Select the **Retain IP** address under IP Config.



Note: The NetApp Shift toolkit supports retaining IP addresses during virtual machine (VM) migrations between different hypervisors, such as from VMware ESXi to Proxmox.



Note: The Shift Toolkit is a powerful solution for VM migration and disk conversion. When preparing migrations, be sure to review and update each VM's resource settings—such as memory and CPU allocation—as these may not be automatically retained during the migration process. Taking the time to verify and adjust these settings ensures optimal performance for your migrated workloads.

16. Click **Continue**.

Virtual Machines Details

Configuration Selection (+)

Service Account (-)

OS	Username	Password	
Linux	root	*****	Apply To All
Windows			Apply To All

IP Config

☐ Do Not Configure ☒ Retain IP ☐ Assign New IP ☐ DHCP

VMs (3)

VM Name	CPU(s)	Mem (MB)	NIC/IP	Power On	Boot Order Override	Gen	Remove VMware Tools	Retain MAC	Service Account Override
Resource Group: RG-vc1									
app1-db	2	1024	192.168.0.63	☒	3	Gen 1 Gen 2	☒	☐	☐
app1-api	2	1024	192.168.0.64	☒	3	Gen 1 Gen 2	☒	☐	☐
app1-web	2	1024	192.168.0.65	☒	3	Gen 1 Gen 2	☒	☐	☐

Figure 2-30:

17. Click **Create Blueprint**.



Note: The NetApp Shift toolkit facilitates scheduled migrations of virtual machines (VMs) between different virtualization environments, primarily from VMware ESXi to Proxmox VE. It allows for both ad-hoc and scheduled migrations, with the latter leveraging a planned maintenance window for minimal

disruption. The toolkit handles disk format conversions and the creation of new VMs on the target Proxmox hypervisor, streamlining the migration process.

The screenshot shows a web interface for 'Schedule Migration'. At the top, there is a navigation bar with four steps: 'Select Resource Groups', 'Set Execution Order', 'Set VM Details', and 'Schedule' (which is the active step, indicated by a blue circle with the number 5). Below the navigation bar, the main heading is 'Schedule Migration'. Underneath, there is a 'Blueprint Details' section. This section contains a table with the following information:

Blueprint Name:	Migrate-vc1
Resource Groups:	Rg-vc1
VMs:	app1-db, app1-api,

Below the table, there is a checkbox labeled 'Schedule' which is currently unchecked. At the bottom of the interface, there are two buttons: 'Previous' and 'Create Blueprint'. The 'Create Blueprint' button is highlighted with a red circle and the number 17, indicating the next step in the process.

Figure 2-31:

18. Once created, a notification will appear at the bottom of the screen stating “Blueprint registered”.



Note: A prepareVM job is initiated automatically, running scripts on source VMs. Ensure VMs are powered on during this phase and that required privileges are set for the preparation scripts. <https://docs.netapp.com/us-en/netapp-solutions-virtualization/migration/shift-toolkit-migrate-esxi2proxmox.html#before-you-begin>

19. Under Status, click **PrepareVM in Progress**.

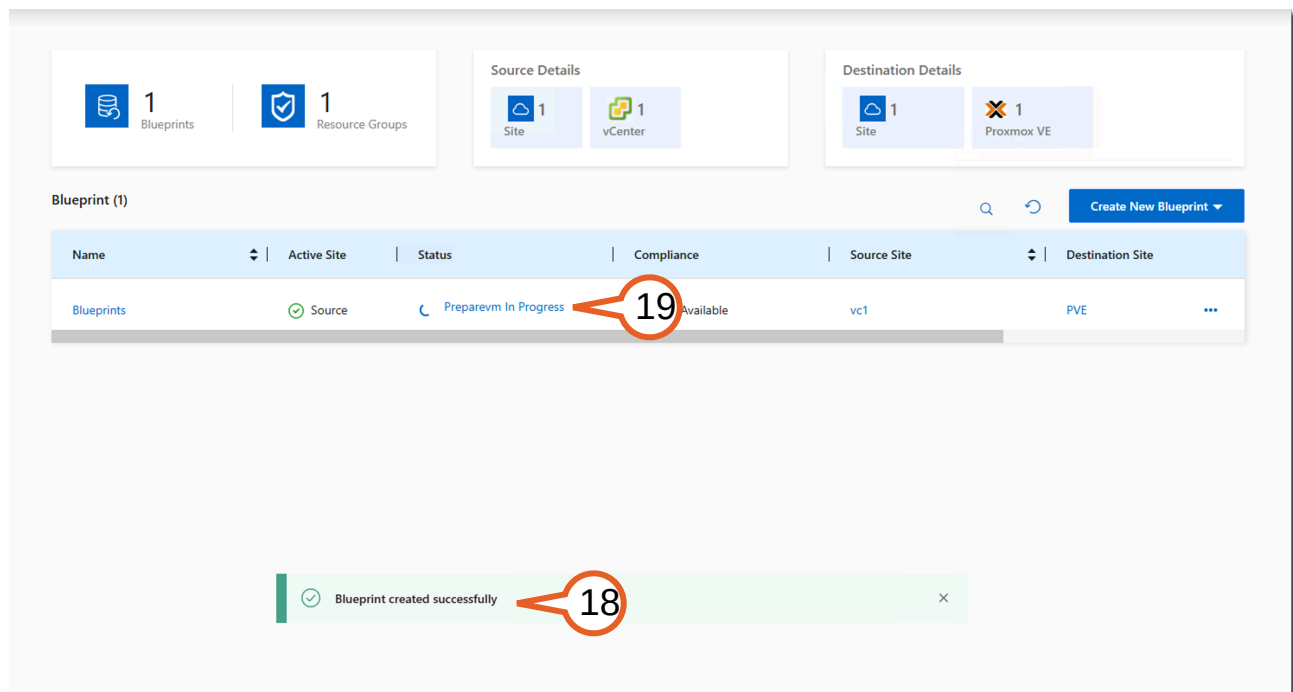


Figure 2-32:

20. Monitor the Prepare VM Steps log.



Note: Prepare VM in the Shift toolkit is crucial because it ensures the virtual machine meets the necessary prerequisites for migration, including installing necessary scripts and drivers, verifying configurations, and addressing potential network interface changes to prevent boot-up or connectivity issues in the destination environment. This pre-migration step helps prevent failures by preparing the

VM's guest operating system for the new hypervisor, allowing for a smoother transition and avoiding unexpected problems with the migrated virtual machine.

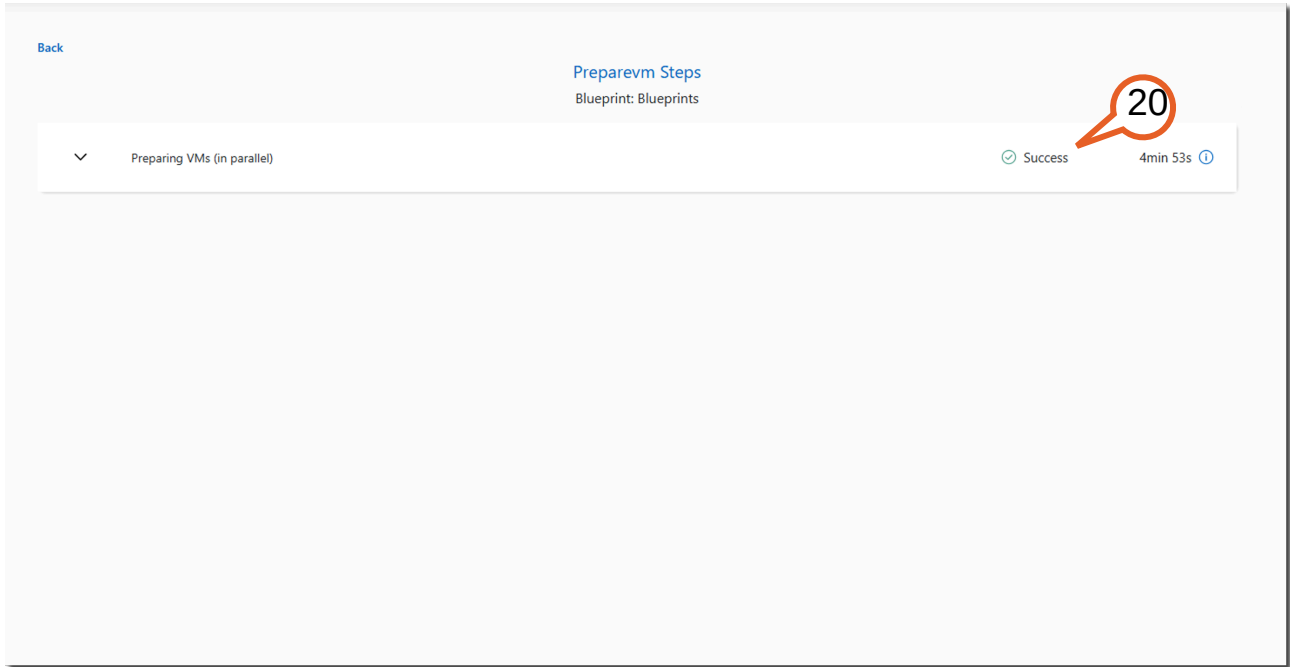


Figure 2-33:

Now that a blueprint is created, you have a comprehensive plan for orchestrating the migration to your virtualized environment, thus ensuring a structured and controlled process to move your VMs between hypervisor platforms.

2.2 Migrate Virtual Machines

Once the blueprint is created, the next step is to migrate the VM. During the migration process, the Shift Toolkit performs a series of steps to convert the disk format (VMDK to QCOW2 or RAW) and uses the converted disk to create virtual machines on the Proxmox VE host as specified in the blueprint.

This section details how to perform VM migration using the Shift Toolkit .

Activities
Connect to Hypervisor Management UIs
Start a VM Migration
Verify the Migration

2.2.1 Connect to Hypervisor Management UIs

In this activity, you login to the source hypervisor and Shutdown VMs before migration.

Before migrating virtual machines with Shift Toolkit, it’s essential to shut them down to ensure a smooth and reliable transition. This practice safeguards data consistency, preserves file integrity, and prevents application errors by ensuring all processes are properly closed. Additionally, performing offline migrations simplifies the workflow and helps prevent potential data loss or corruption, especially when moving between different hypervisor platforms like VMware and Proxmox.

- 1. Click **vCenter** on the bookmark browser.

2. Click **LAUNCH VSPHERE CLIENT**.

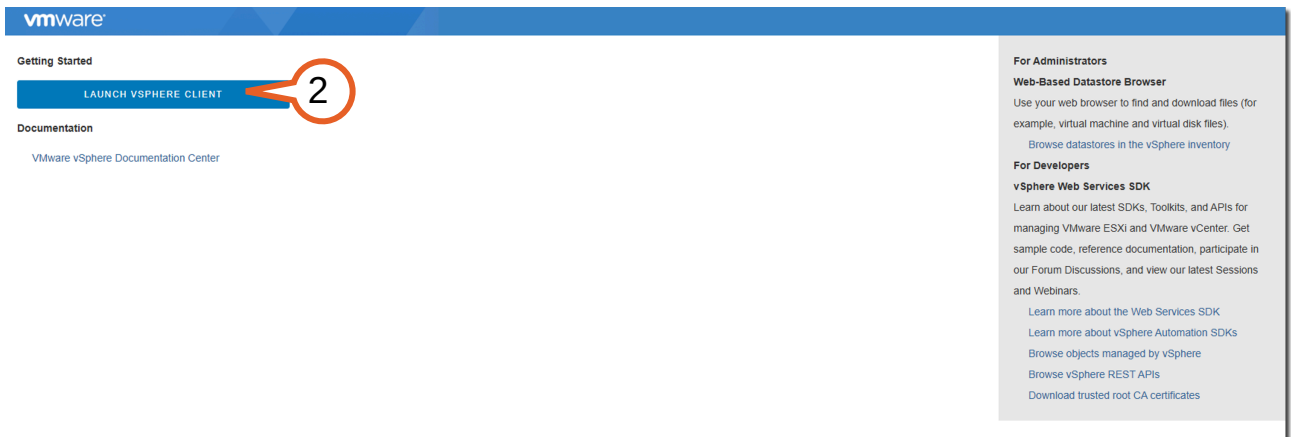


Figure 2-34:

3. Add the following VMware vSphere Credentials:

- Username: **administrator@demo.local**
- Password **Netapp1!**

4. Click **LOGIN**.



Figure 2-35:

5. Verify the following vm's are Powered on:

- app1-api
- app1-db
- app1-web

6. Right click on each of the following VMs and select Power->Shutdown Guest OS. (combine steps):

- **app1-api**
- **app1-db**
- **app1-web**

7. Navigate to **Power** and **Shutdown Guest OS**.

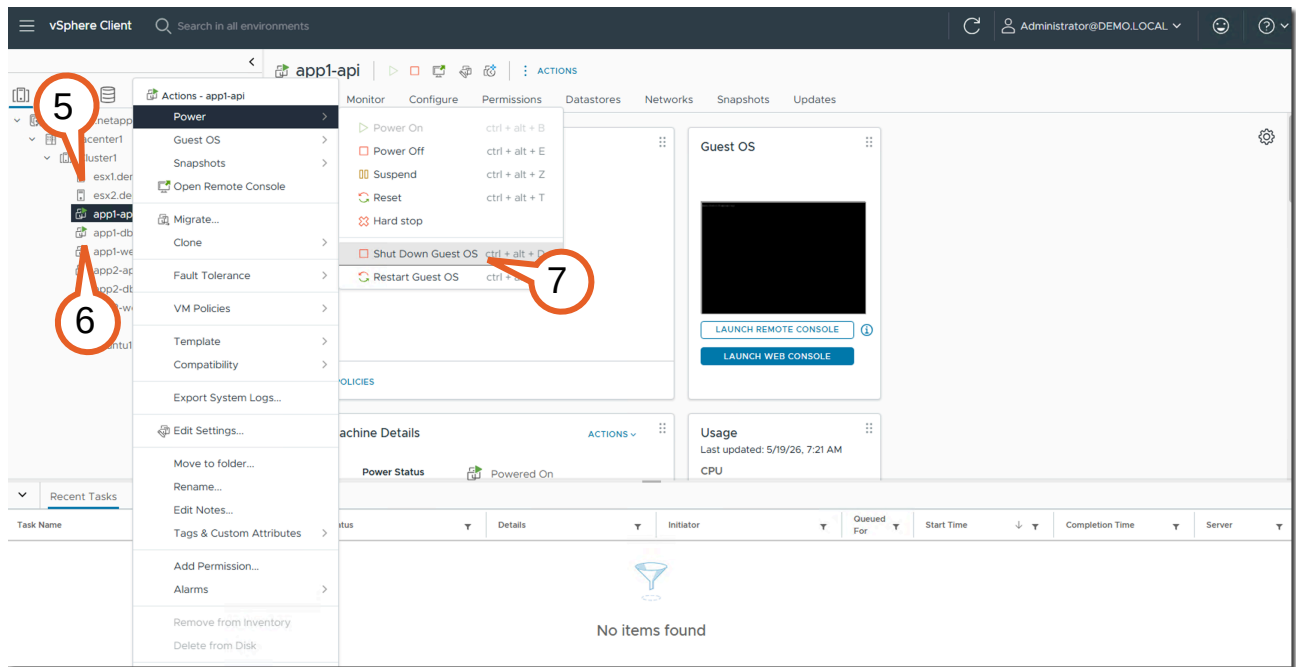


Figure 2-36:

8. Click **YES**.

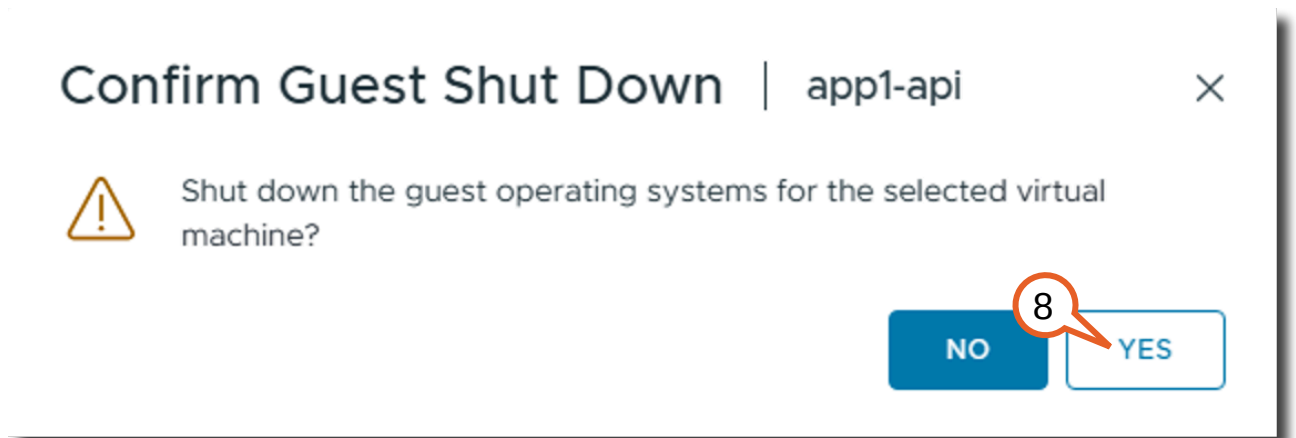


Figure 2-37:

By shutting down VMs before migration with Shift Toolkit, we ensure a safe, consistent, and reliable move to the new environment, reducing the risk of data loss or corruption. With this step complete, we're now ready to begin the VM migration process.

2.2.2 Start a VM Migration

In this activity, initiate the migration of the VMs that were powered off in the section above..

1. Once virtual machine is powered on navigate back to the **Shift** tab.



Note: Clone based Migration – performs end to end migration of the VM from source hypervisor to destination hypervisor.

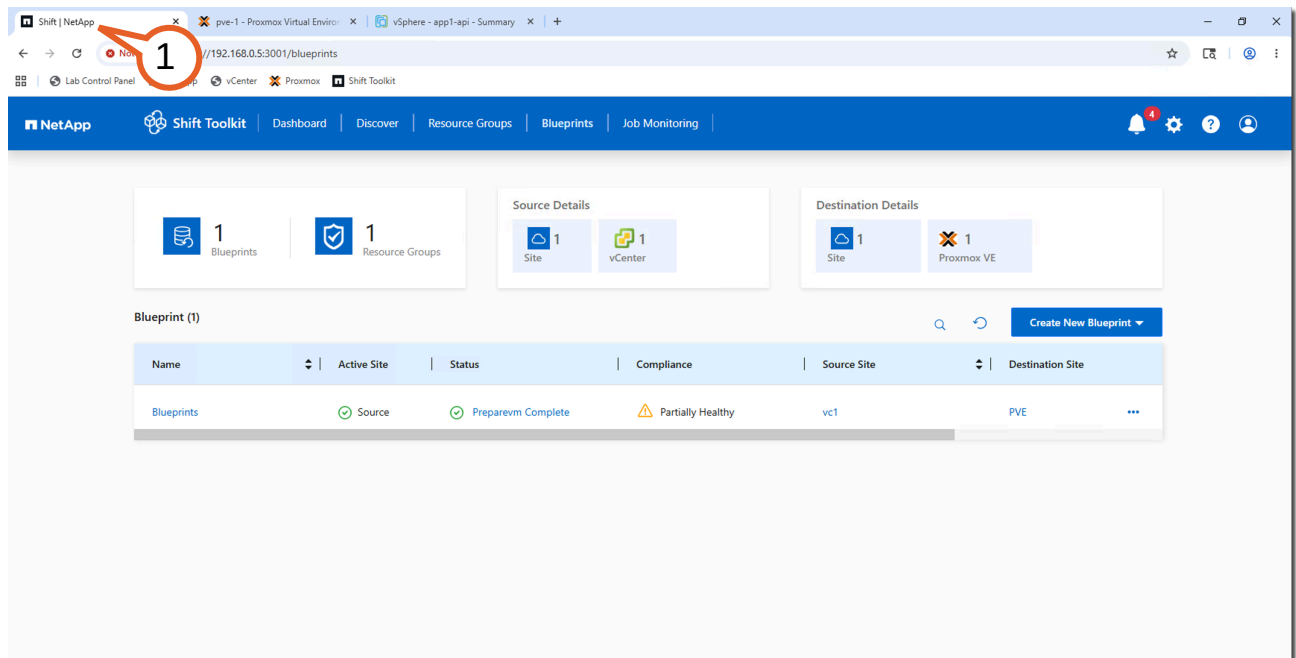


Figure 2-38:

2. Jump over to the **Dashboard**.



Note: The VM is listed as “Yet to Move”, and if you have a large number of the VMs to migrate the dashboard can help track the overall status of all the work to be performed.

3. Click **Blueprints** to kick off the migration.

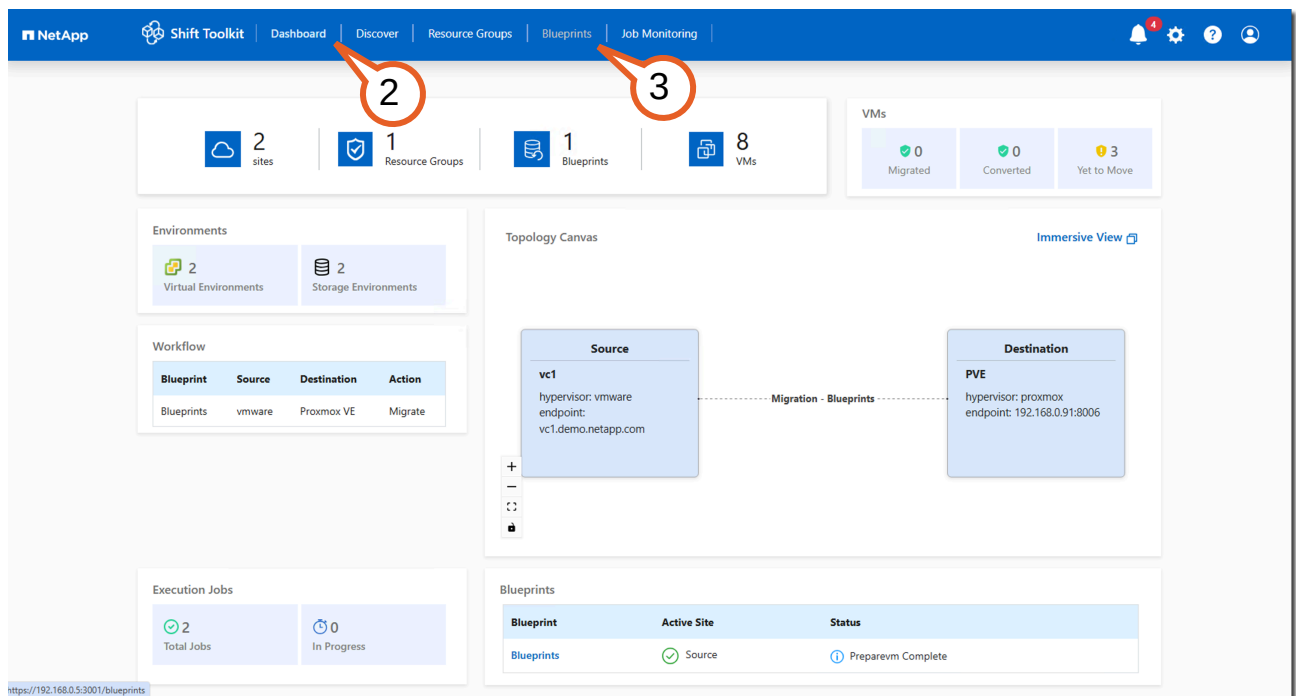


Figure 2-39:

4. On the far right, click the **three dots** menu.
5. Click **Migrate**.



Tip: You can also test a migration to ensure it will work as expected, particularly when you have a scheduled plan and want to confirm it will complete successfully. The Test Migrate feature simulates a full migration by converting virtual disk files (VMDK to QCOW2, or RAW for VMware to Proxmox VE) and creating a new VM on the destination Proxmox VE environment using the converted disks. This allows you

to validate the migration process end-to-end without impacting your production environment, giving you confidence that your scheduled migration will proceed as planned.

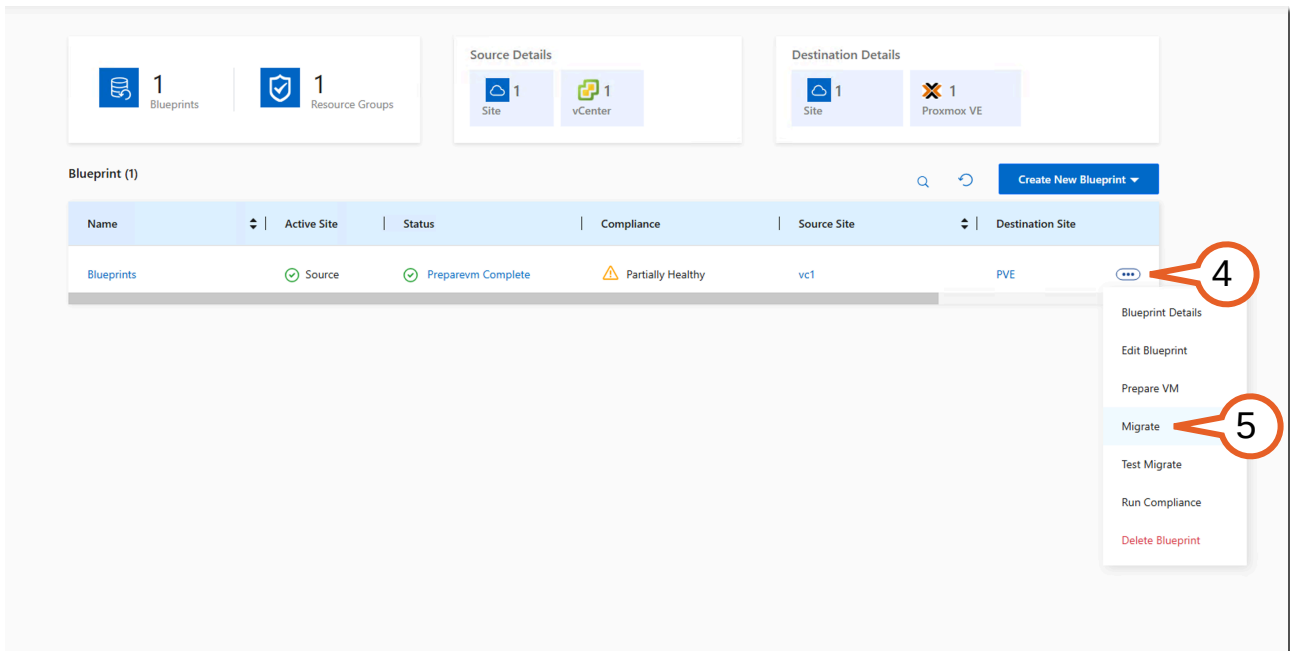


Figure 2-40:

6. During migration, you will be directed to the Migrate Steps page, where you can monitor progress and any issues that might arise. The rapid migration speeds you see are made possible by ONTAP’s FlexClone technology in

step 4, which creates space-efficient snapshots, and Shift Toolkit’s in-place virtual disk conversion in step 7. This combination enables fast, efficient migrations with minimal downtime..

Note: If the VMs within the blueprint are not powered off, the Shift toolkit will prompt for a graceful shutdown before proceeding.

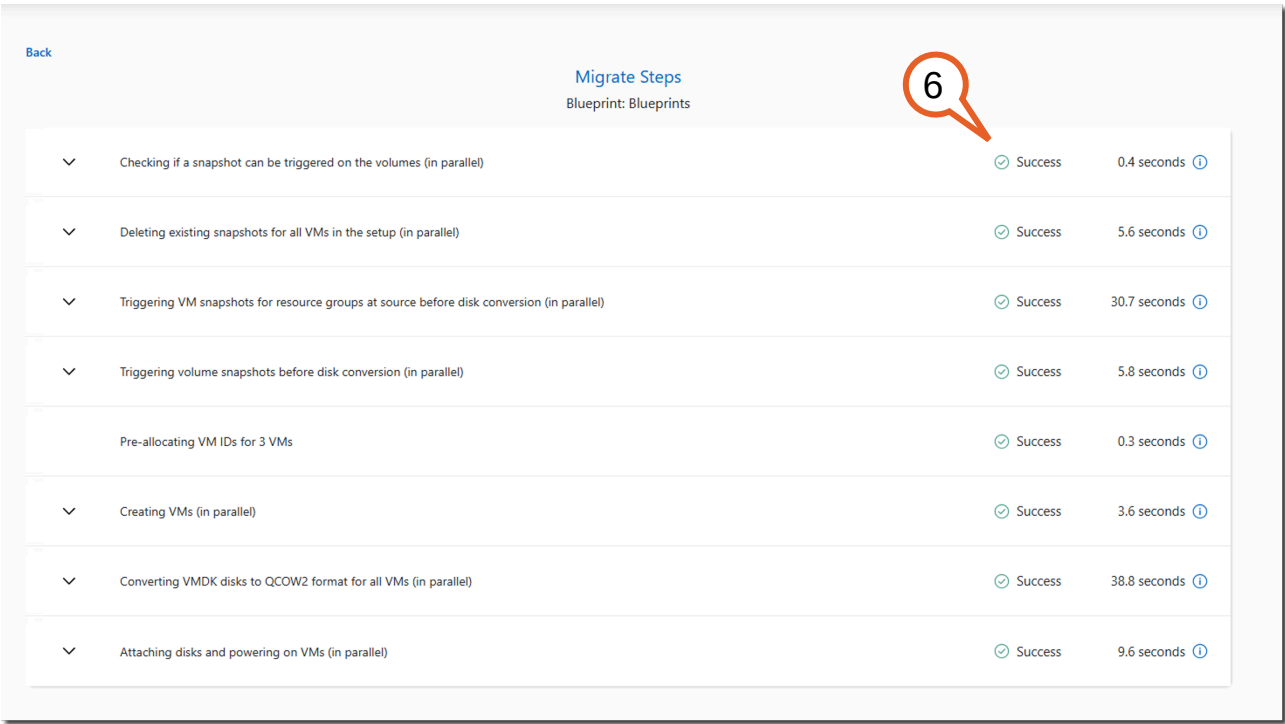


Figure 2-41:

You have successfully migrated a VM from one hypervisor platform to another.

2.2.3 Verify the Migration

Now that the migration has successfully migrated to the destination hypervisor, you need to verify that it is working.

1. Click the **Google Chrome** browser icon on Desktop.

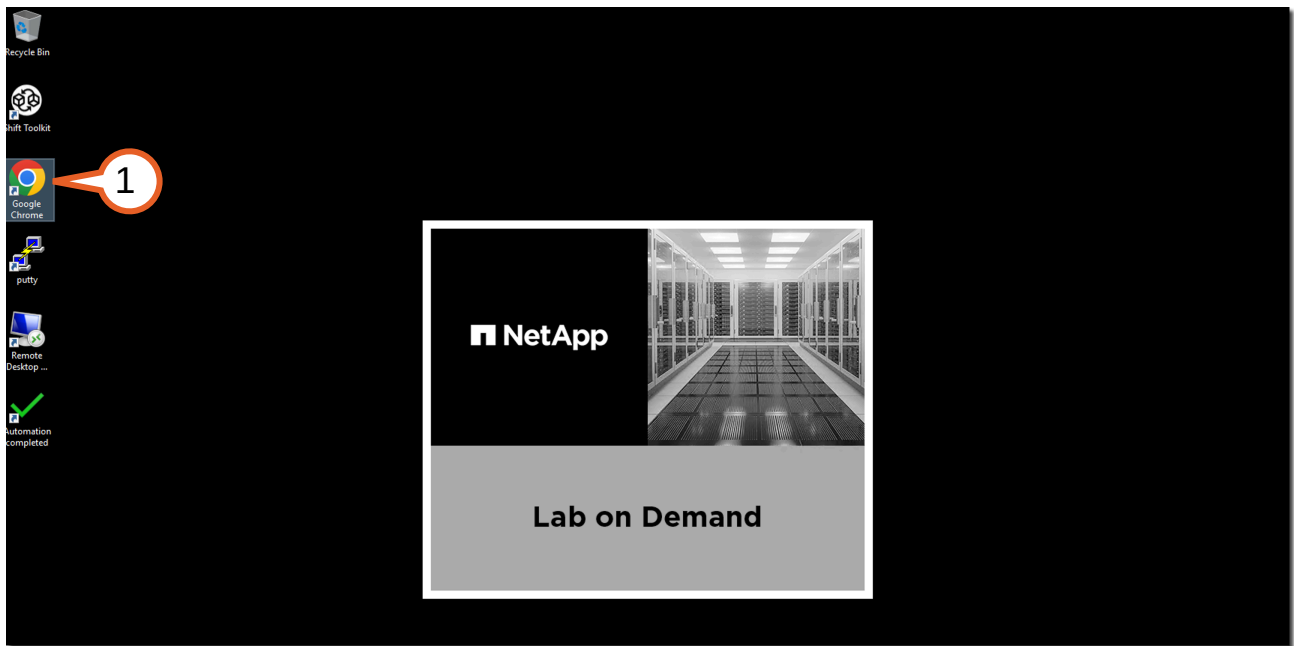


Figure 2-42:

2. Click **Proxmox** that is saved in the bookmarks.

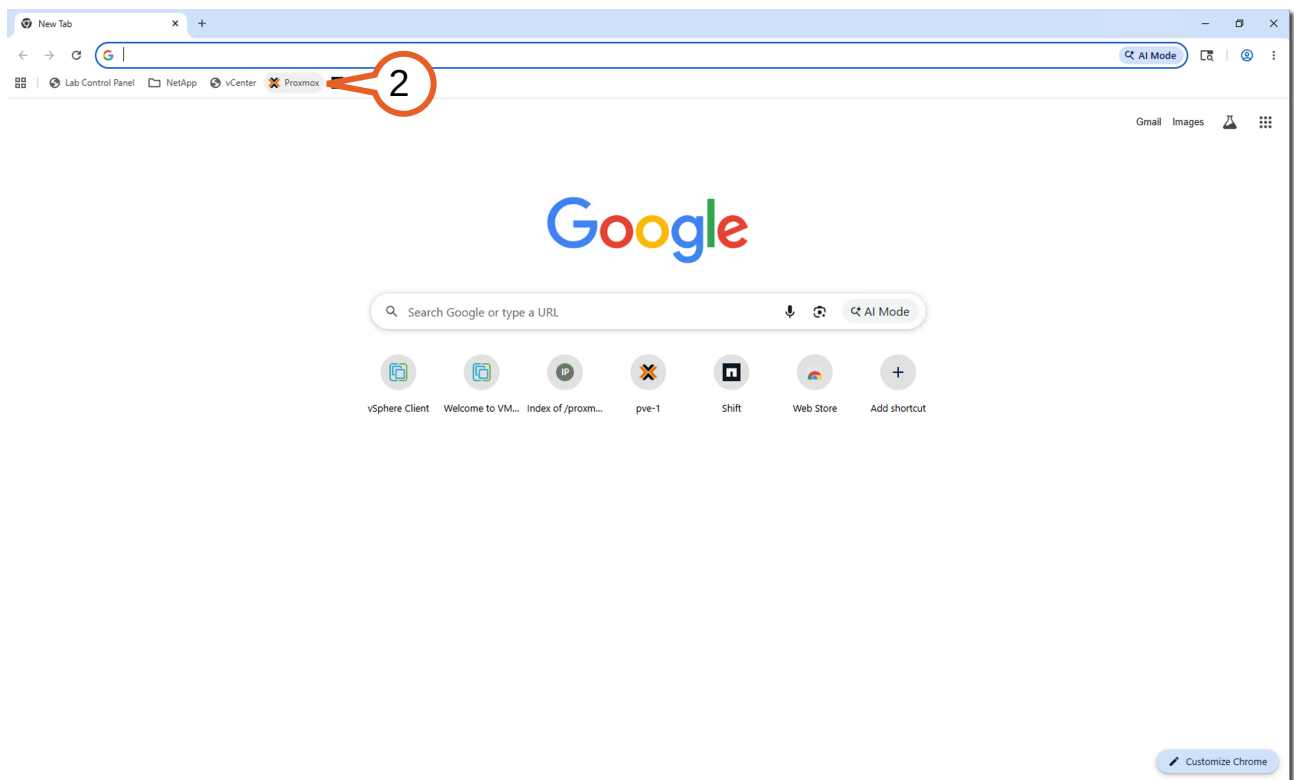


Figure 2-43:

3. Login into Proxmox VE with the following credentials:

- Login: **root**
- Password: **Netapp1!**

4. Click **Login**.

The image shows the Proxmox VE Login interface. A red box labeled '3' highlights the 'User name' field containing 'root' and the 'Password' field containing '.....'. A red box labeled '4' highlights the 'Login' button. Other fields include 'Realm' set to 'Linux PAM standard authentication' and 'Language' set to 'English - English'. A 'Save User name' checkbox is checked.

Proxmox VE Login

User name: **root**

Password: **.....**

Realm: **Linux PAM standard authentication**

Language: **English - English**

Save User name: ☒ **Login**

Figure 2-44:

5. Click to expand **pve-1**.

The image shows the Proxmox VE Datacenter view. A red box labeled '5' highlights the 'pve-1' node in the left sidebar. The main table displays system resources for pve-1 and pve-2.

PROXMOX Virtual Environment 9.1.5

Server View | Datacenter

Search

Search

Type	Description	Disk usage...	Memory us...	CPU usage	Uptime	Host CPU ...	Host Mem ...	Tags
zone	localnetwork (pve-1)				-			
zone	localnetwork (pve-2)				-			
node	pve-1	19.8 %	28.9 %	4.3% of 4 ...	02:30:11			
node	pve-2	19.7 %	15.7 %	1.6% of 4 ...	02:30:09			
qemu	100 (ubuntu)				-			
qemu	101 (app1-web)	0.0 %	34.0 %	0.7% of 2 ...	00:29:58	0.4% of 4 ...	4.2 %	
qemu	102 (app1-api)	0.0 %	33.8 %	0.8% of 2 ...	00:29:57	0.4% of 4 ...	4.2 %	
qemu	103 (app1-db)	0.0 %	33.6 %	0.8% of 2 ...	00:29:56	0.4% of 4 ...	4.1 %	
storage	iscsi1 (pve-1)				-			
storage	iscsi1-lvm (pve-1)	0.0 %			-			
storage	local (pve-1)	19.8 %			-			
storage	local-lvm (pve-1)	0.0 %			-			
storage	nfs1 (pve-1)	81.8 %			-			
storage	nfs1-pve (pve-1)	81.8 %			-			
storage	iscsi1 (pve-2)				-			
storage	iscsi1-lvm (pve-2)	0.5 %			-			
storage	local (pve-2)	19.7 %			-			
storage	local-lvm (pve-2)	0.5 %			-			
storage	nfs1 (pve-2)	81.8 %			-			
storage	nfs1-pve (pve-2)	81.8 %			-			

Figure 2-45:

6. Click **app1-web**.

7. Verify the IP address for app1-web1 is retained post migration in the summary.

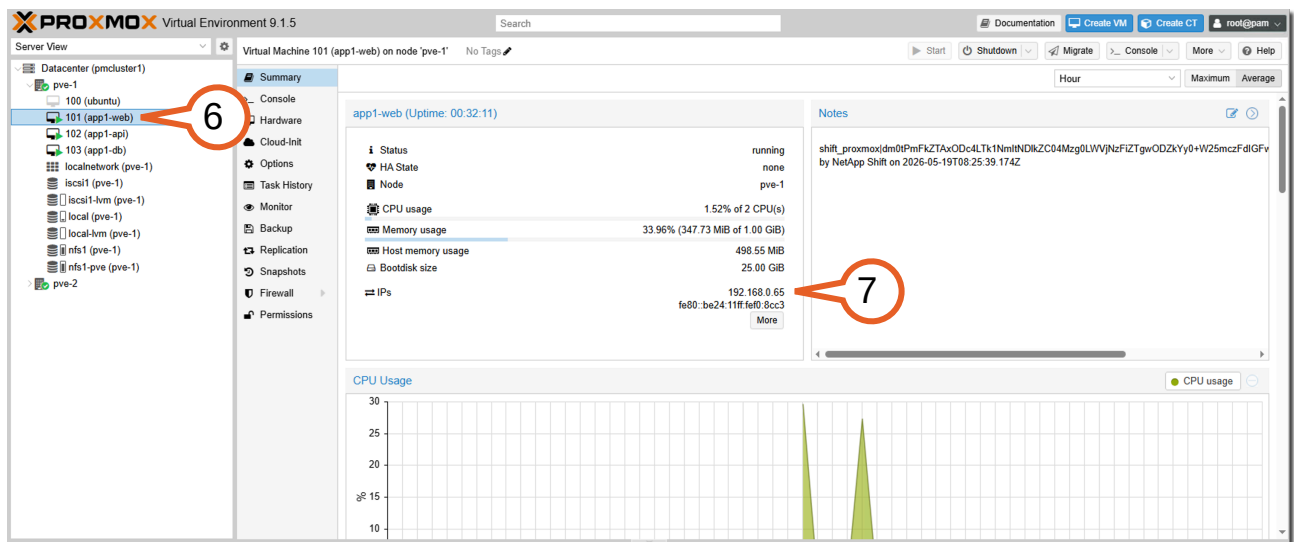


Figure 2-46:

This activity showed you how to use the NetApp Shift toolkit to perform a migration by converting the disks, then adding the converted VM to a new datastore for the destination hypervisor.

3 Lab Summary

The NetApp Shift toolkit helps administrators rapidly and seamlessly convert VMs from VMware to Proxmox VE. It can also convert just the virtual disks between different hypervisors. The Shift toolkit can save you several hours of effort each time you want to move workloads from one hypervisor to another. Organizations can now host multi-hypervisor environments without having to worry about workloads being tied to a single hypervisor. This capability increases flexibility and reduces licensing costs, vendor lock-in, and long-term commitments to a single platform.

NetApp Shift Toolkit supports **disk-level conversion** for these virtual disk formats (i.e., source # target):

- - VMDK # VHDX (VMware ESXi to Microsoft Hyper-V)
 - VHDX # VMDK (Microsoft Hyper-V to VMware ESXi)
 - VMDK # QCOW2 (VMware ESXi to KVM-compatible hypervisors)
 - VMDK # RAW (VMware ESXi to KVM-compatible hypervisors)

4 Additional Information

4.1 Lab Environment

All of the servers and storage controllers presented in this lab are virtual devices, and the networks that interconnect them are exclusive to your lab session. While the virtual storage controllers (vsims) used in this lab offer nearly all of the same functionality as physical storage controllers, they are not capable of providing the same performance as a physical controller, so they are not suitable for performance testing.

The Lab Host Credentials table provides a list of the servers and storage controller nodes in the lab, along with their IP addresses.

Table 1: Lab Host Credentials

Hostname	Description	IP Address(es)	Username	Password
Jumphost	Primary desktop entry point for lab	192.168.0.5	administrator@demo.netapp1	Netapp1!
dc1	Active Directory / DNS	192.168.0.253	administrator@demo.netapp1	Netapp1!
Win2	Secondary desktop with Hyper-v	192.168.0.41	administrator@demo.netapp1	Netapp1!
Vc1	vCenter	192.168.0.31	administrator@demo.netapp1	Netapp1!
PVE(192.168.0.91:8006)	Proxmox VE	192.168.0.91:8006	root	Netapp1!
Cluster1	ONTAP	192.168.0.111	admin	Netapp1!
Cluster2	ONTAP	192.168.0.112	admin	Netapp1!

4.2 References

The following references were used to write this lab guide:

- [Learn about migrating VMs between virtualization environments using NetApp Shift Toolkit](#)
- [Migrate VMs using the Shift Toolkit](#)
- [Convert VMs using the Shift Toolkit](#)
- [Next steps after converting or migrating VMs using the Shift Toolkit](#)
- <https://docs.netapp.com/us-en/netapp-solutions-virtualization/migration/shift-toolkit-settings.html>

4.3 Version History

Version	Date	Document Version History
1.0	May 2026	Initial Release.

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